Model Derivations

Companion notes to "Trade Tariffs and Exchange Rates"

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These notes build the model once, for generic N, and then specialize: environment and definitions (Sections 1–3), the general linear system (Section 4), two countries (Section 5), three countries (Section 6), general symmetric N (Section 7), asymmetry (Section 8), and the canonical PTA model as a parameter restriction (Section 9). Every closed form is verified numerically in the replication package. Prose is limited to defining variables and linking equations.

Notation. Countries $i, j \in \{1, \ldots, N\}$; A is the tariff imposer, B the target, C a bystander. e_{ij} = units of i's currency per unit of j's currency, so a *rise* in e_{ij} is a depreciation of i against j; $e_{ii} = 1$ and $e_{ij} = e_{ik}e_{kj}$. Perturbations: $\nu_j \equiv d\log e_{Aj}$ with $\nu_A = 0$ (A's currency is the numeraire). Tariffs $\tau_{ij} \geq 0$: ad valorem, levied by i on imports from j, revenue rebated lump-sum. CES weights enter shares linearly (share-linear convention). Recurring composites:

$$m \equiv \alpha_T(1-\alpha_D), \qquad \rho_1^* \equiv 1+\alpha_D(\eta-1)-\alpha_T(1-\alpha_D), \qquad D_N \equiv N\alpha_D(\eta-1)+(N-2)\rho+1. \quad (1)$$

1 Environment

1.1 Technology

Each country produces a tradable T_i (only i produces it, all countries consume it) and a non-tradable N_i, from labor alone:

$$Y_{T_i} = A_{T_i}L_{T_i}, \qquad Y_{N_i} = A_{N_i}L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i. \quad (2)$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price in i's own currency) give

$$w_i = A_{T_i}, \qquad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \quad (3)$$

Wages are constant in own currency; all adjustment runs through exchange rates (Section 2.5).

1.2 Preferences: two-layer nest

$$C_i = C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} \qquad \text{(Cobb–Douglas outer, tradable share } \alpha_T\text{)} \tag{4}$$

$$C_{T_i} = \left[\alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}} \qquad \text{(home vs. imports, elasticity } \eta,\ \text{home weight } \alpha_D\text{)} \tag{5}$$

$$M_i = \left[\ \textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \qquad \text{(across origins, elasticity } \rho,\ \text{weights } \textstyle\sum_{j\neq i}\beta_{ij}=1\text{)} \tag{6}$$

The single-elasticity CES is the special case $\rho = \eta$; Cobb–Douglas within tradables is $\eta = 1$.

Symmetry operates at three levels, and the symmetric benchmark used below imposes all three: equal sizes and technologies ($L_i = L,\ A_{T_i} = A_T,\ A_{N_i} = A_N$), common preference parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal origin weights $\beta_{ij} = 1/(N-1)$. Home and foreign tradables remain asymmetric through the home weight α_D: full symmetry across all N tradables would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which is not imposed.

1.3 Policy and closure

The consumer price of T_j in i is $p_{ij} = (1+\tau_{ij})\, e_{ij}$ (in i's currency). Nothing requires $\tau_{ii} = 0$: a tax on home tradables ($p_{ii} = 1+\tau_{ii}$, rebated like the rest) fits the same structure. We set $\tau_{ii} = 0$ throughout, so that τ is a border instrument and $p_{ii} = 1$. Revenue is rebated lump-sum (Section 3.2).

Assumption 1 (Closure)**.** *There are no internationally traded assets: each country's trade balance is zero every period,* $\mathrm{TB}_i = 0$ *for all* i*.*

Assumption 1 is the margin that pins down exchange rates.

2 Prices, incomes, and exchange-rate concepts

2.1 Price indices

CES price indices in i's currency, from (5)–(6):

$$P_{M_i} = \left[\textstyle\sum_{j\neq i} \beta_{ij}\, p_{ij}^{1-\rho}\right]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \left[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta}\right]^{\frac{1}{1-\eta}}, \tag{7}$$

and the consumer price index from (4) (κ_i collects share constants):

$$P_i = \kappa_i\, P_{T_i}^{\alpha_T}\, P_{N_i}^{1-\alpha_T}. \tag{8}$$

2.2 Definitions: the objects to be signed

$$\omega_{ij} \equiv \frac{e_{ij} w_j}{w_i} = \frac{A_{T_j}}{A_{T_i}} e_{ij} \;\propto\; e_{ij} \qquad \text{(relative wage, common currency)} \tag{9}$$

$$\mathrm{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij} P_{T_j}} = \frac{1}{e_{ij}} \qquad \text{(bilateral terms of trade, pre-tariff prices)} \tag{10}$$

$$q_{ij} \equiv \frac{e_{ij} P_j}{P_i} \qquad \text{(bilateral real exchange rate)} \tag{11}$$

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij}\, d\log e_{ij}, \qquad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i}(f_{ik} + f_{ki})} \qquad \text{(NEER, trade weights)} \tag{12}$$

$$dq_i^E \equiv \sum_{j \neq i} w_{ij}\, d\log q_{ij} \qquad \text{(REER, same weights)} \tag{13}$$

where f_{ij} is the border value of i's imports from j in a common currency — the same flows that enter the trade balances, defined formally in (18) below (the currency choice cancels from the weights). All effective rates are depreciation-positive: $d\nu_i^E > 0$ means i's currency loses value on average. In the symmetric model $w_{ij} = 1/(N-1)$.

2.3 Wage–exchange-rate equivalence

Prices and demands depend on w_j and e_{ij} only through the products $e_{ij} w_j$. The model therefore does not separate relative wage adjustment from nominal exchange-rate adjustment: fixing $w_i = A_{T_i}$ and letting e adjust (as here) is equivalent to fixing e and letting relative wages adjust. Moreover, with one factor and constant returns, producer prices are unit labor costs, so (9)–(10) give

$$d\log e_{ij} \;=\; d\log \omega_{ij} \;=\; -\,d\log \mathrm{ToT}_{ij} : \tag{14}$$

the nominal exchange rate, the common-currency relative wage, and the (inverse) terms of trade move one-for-one. Every result below about e_{ij} is simultaneously a statement about all three.

3 Demands and equilibrium

3.1 Expenditure shares

With income I_i, the share-linear CES demands from (4)–(6) give expenditure shares of income (rows sum to α_T over tradables):

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}}\right)^{1-\eta}, \tag{15}$$

$$s_{ij} = \alpha_T\, (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}}\right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}}\right)^{1-\rho}, \qquad j \neq i. \tag{16}$$

Shares are tariff-inclusive: $s_{ij}I_i$ is spending at consumer prices; $s_{ij}I_i/(1+\tau_{ij})$ is the producer-price (border) value.

3.2 Income

A fraction $\tau_{ij}/(1+\tau_{ij})$ of spending $s_{ij}I_i$ is tariff revenue, rebated:

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad \Longrightarrow \quad I_i = \frac{w_i L_i}{1-\sum_{j\neq i}\frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{17}$$

The shares depend only on prices, so (17) is a closed form.

3.3 Trade balances and equilibrium

In a common currency, at producer prices (the tariff wedge is a domestic transfer),

$$\mathrm{TB}_i = \underbrace{\sum_{k\neq i} f_{ki}}_{\text{exports}} - \underbrace{\sum_{j\neq i} f_{ij}}_{\text{imports}}, \qquad f_{ij} \equiv \frac{s_{ij}\, I_i}{1+\tau_{ij}} \quad (i\text{'s imports from } j\text{, border value}). \tag{18}$$

By Walras' law $\sum_i \mathrm{TB}_i = 0$: only $N-1$ conditions are independent, matching the $N-1$ free exchange rates. Equilibrium: given $\{\tau_{ij}\}$, the rates $\{e_{Aj}\}_{j\neq A}$ solve $\mathrm{TB}_i = 0$ for $i \neq A$.

4 The general linear system

4.1 Linearization at a balanced free-trade baseline

Evaluate at $\tau_{ij} = 0$ with balanced trade, and normalize baseline incomes. Baseline objects: home share within tradables $h_i = \alpha_D$ (symmetric weights case: general h_i carries through identically), within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = \alpha_T(1-\alpha_D)b_{ij} = m\, b_{ij}$, and flow values $f_{ij} = s_{ij}I_i$ (from (18) at $\tau = 0$). Balance requires $\sum_k f_{ki} = \sum_j f_{ij}$.

Perturb by tariffs dt_{ij} and collect exchange-rate changes $\nu_j = d\log e_{Aj}$. Log price changes in i's own currency ($dp_{ii} = 0$):

$$dp_{ij} = \nu_j - \nu_i + dt_{ij}. \tag{19}$$

Index changes, from (7):

$$d\log P_{M_i} = \sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{T_i} = (1-\alpha_D)\, d\log P_{M_i}. \tag{20}$$

Share changes, from (15)–(16), using $d\log P_{M_i} - d\log P_{T_i} = \alpha_D\, d\log P_{M_i}$:

$$d\log s_{ii} = -(1-\eta)(1-\alpha_D)\, d\log P_{M_i}, \tag{21}$$

$$d\log s_{ij} = \underbrace{(1-\rho)\big(dp_{ij} - d\log P_{M_i}\big)}_{\text{across origins}} + \underbrace{(1-\eta)\,\alpha_D\, d\log P_{M_i}}_{\text{home vs. imports}}. \tag{22}$$

Income changes in the common currency, from (17) (revenue is first order only for tariffing countries):

$$d\log I_i = \nu_i + \sum_{j\neq i} s_{ij}\, dt_{ij}. \tag{23}$$

Differentiating (18), each flow contributes value × log change:

$$d\,\mathrm{TB}_i = \sum_{k\neq i} f_{ki}\big(d\log s_{ki} + d\log I_k - dt_{ki}\big) - \sum_{j\neq i} f_{ij}\big(d\log s_{ij} + d\log I_i - dt_{ij}\big). \tag{24}$$

Equations (19)–(24) are linear in the exchange-rate changes and the tariff changes. Fix a policy direction $a = (a_{jk})$ and scale it by a single number $d\tau$: the experiment is $dt_{jk} = a_{jk}\, d\tau$ (the unilateral tariff of the sections below is $a_{AB} = 1$, all other $a_{jk} = 0$; the trade war is $a_{AB} = a_{BA} = 1$). Stack the free rates in the vector $d\nu \equiv (\nu_j)_{j\neq A} \in \mathbb{R}^{N-1}$ and define

$$J_{il} \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial \nu_l}, \qquad F_i \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial\, d\tau}\bigg|_{\nu=0} = \sum_{j,\,k} a_{jk}\, \frac{\partial\, d\mathrm{TB}_i}{\partial\, dt_{jk}}\bigg|_{\nu=0}, \qquad i, l \neq A, \tag{25}$$

both read off (24): J collects the coefficients on ν, and F collects the coefficients on $d\tau$ at unchanged exchange rates. The equilibrium conditions $d\mathrm{TB}_i = 0$, $i \neq A$, then read

$$J\, d\nu = -F\, d\tau \quad\Longrightarrow\quad \frac{d\nu}{d\tau} = (-J)^{-1}F = \frac{\mathrm{adj}(-J)\, F}{\det(-J)}, \tag{26}$$

where $\mathrm{adj}(-J)$ is the adjugate (transpose of the cofactor matrix). J is the adjustment mechanism (how balances respond to exchange rates); F is tariff incidence (how the tariff disturbs balances at unchanged rates). The entries of both are linear in η and ρ with baseline shares as coefficients.

4.2 Properties of J: the multilateral Marshall–Lerner condition

Write $\widetilde{J}$ for the full $N \times N$ Jacobian, $\widetilde{J}_{il} = \partial\, d\mathrm{TB}_i/\partial\nu_l$ for all $i, l \in \{1, \ldots, N\}$, before dropping country A: the J of (25) is $\widetilde{J}$ with row and column A deleted. Two properties of $\widetilde{J}$ are identities of the model, not assumptions:

$$\sum_l \widetilde{J}_{il} = 0 \quad \forall i \qquad \text{(homogeneity: (19) depends only on differences } \nu_j - \nu_i) \tag{27}$$

$$\sum_i \widetilde{J}_{il} = 0 \quad \forall l \qquad \text{(Walras' law: } \textstyle\sum_i \mathrm{TB}_i \equiv 0). \tag{28}$$

Assumption 2 (Gross substitutability)**.** *$\widetilde{J}_{il} \geq 0$ for all $l \neq i$: a depreciation of any country l (a rise in ν_l) does not worsen any other country i's trade balance.*

The condition is on every off-diagonal entry — all country pairs, not only A's partners. In the explicit symmetric systems of Sections 6–7, every off-diagonal entry is a positive multiple of D_N, so Assumption 2 there is exactly $D_N > 0$ and holds for all $\eta \geq 1$.

Combining (27) with Assumption 2 gives the diagonal, and then the reduced system inherits a dominant diagonal:

$$\widetilde{J}_{ii} = -\sum_{l\neq i} \widetilde{J}_{il} \;\le\; 0, \qquad\qquad |J_{ii}| = \sum_{l\neq i,\,A} J_{il} \;+\; \widetilde{J}_{iA} \;\ge\; \sum_{l\neq i,\,A} J_{il}. \tag{29}$$

The first equation is the set of bilateral Marshall–Lerner conditions, one per country: own depreciation improves own balance. The second says each row of $-J$ dominates its off-diagonal sum, strictly in every row i with $\widetilde{J}_{iA} > 0$ (every country whose balance responds to A's rate). So diagonal dominance is not an independent assumption and not merely a restatement of the identities: it follows from homogeneity (27) *plus* gross substitutability, once the numeraire column is dropped. Walras' law (28) delivers the same dominance columnwise.

Result 4.1 (Multilateral Marshall–Lerner condition)**.** *Under Assumption 2, if J is irreducible (a connected trade network) with $\widetilde{J}_{iA} > 0$ for some i, then $-J$ is a nonsingular M-matrix: a matrix with nonpositive off-diagonal entries satisfying the equivalent conditions*

$$\det(-J) > 0 \textit{ (all principal minors positive)}, \qquad (-J)^{-1} \ge 0 \textit{ elementwise}, \qquad \operatorname{Re}\lambda(J) < 0. \tag{30}$$

We refer to (30) *as the* multilateral Marshall–Lerner condition*.*

The proof is the standard characterization of irreducibly diagonally dominant Z-matrices; (29) supplies the dominance. Three clarifications of the logic:

- Gross substitutability is a *sufficient primitive*, not the condition itself: Assumption 2 implies (30), but (30) can hold without it. Everything below uses only (30).
- The eigenvalue statement in (30) is local stability of the price-adjustment process $\dot{\nu}_i = k_i\,\mathrm{TB}_i$ with any speeds $k_i > 0$ (the *tatonnement*: a country in surplus sees its currency appreciate). Stability is thus not an extra requirement layered on top of (30) — it is one of its equivalent faces, and M-matrices are stable for every choice of speeds.
- At $N = 2$, all of (30) collapses to the single inequality $J < 0$, whose classical elasticity form $\varepsilon_X + \varepsilon_M > 1$ is derived in Section 5.3.

4.3 The sign rule

With J and F as defined in (25), the solution $d\nu/d\tau = (-J)^{-1}F$ combines the impact effects through a nonnegative matrix:

Result 4.2 (Sign rule)**.** *Under the multilateral Marshall–Lerner condition* (30)*,*

$$\operatorname{sign}\!\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\Big(\sum_j \omega_{ij} F_j\Big), \qquad \omega_{ij} \equiv \big[(-J)^{-1}\big]_{ij} \ge 0. \tag{31}$$

Every response is a nonnegative combination of the impact effects. If all F_j share one sign, every exchange rate moves the same way; opposite-signed responses require an impact vector with mixed signs.

For the discriminatory tariff studied below, the impact vector is generically mixed, so the sign rule has content. At fixed exchange rates the tariff always worsens the target's balance ($F_B < 0$: B loses export demand to A), while a bystander's balance combines a gain (demand diverted from B, increasing in ρ) with a loss (sales to a poorer B): in the symmetric three-country case $F_C \propto \rho - \rho_1^*$ (Section 6.1), positive exactly when diversion dominates. The thresholds derived below are the points where a weighted impact sum $\sum_j \omega_{ij} F_j$ crosses zero. Because F is linear in ρ and $\mathrm{adj}(-J)$ has degree $N-2$ in ρ, each such sum is a polynomial of degree $N-1$ in ρ: one relevant root at $N=3$ (Section 6), possibly several sign changes at $N \geq 4$.

5 Two countries

5.1 The slope

At $N=2$ there is one balance and one rate. Specializing (19)–(24) (details: Section 7 at $N=2$; ρ is inoperative with a single origin):

$$\boxed{\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_1^*}{D_2}, \qquad D_2 = 2\alpha_D(\eta-1)+1.} \tag{32}$$

The numerator is the object defined in (1), with the identity

$$\rho_1^* = 1+\alpha_D(\eta-1) - \alpha_T(1-\alpha_D) = \underbrace{\alpha_D\,\eta}_{\text{home switching}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{expenditure absorption}} > 0 \tag{33}$$

for all admissible parameters.

Result 5.1 (Conventional wisdom, $N=2$)**.** *Under the multilateral Marshall–Lerner condition* (30) *— which at $N=2$ is the single inequality $J<0$, equivalently $D_2>0$, and which Section 5.3 shows is exactly the textbook condition $\varepsilon_X+\varepsilon_M>1$ — a tariff appreciates the tariffing country's currency unconditionally, since $\rho_1^*>0$ by* (33)*. The condition holds for all $\eta \geq 1$ and fails only under strong complementarity, $\eta < 1-1/(2\alpha_D)$.*

5.2 Exact solution in the Cobb–Douglas case

At $\eta=1$ shares are constant: $s_{AB}=s_{BA}=m$. From (17), $I_A = w_A L_A/\big[1-\frac{\tau}{1+\tau}m\big]$, and $\mathrm{TB}_A=0$ in (18) reads $e\,m\,w_B L_B = mI_A/(1+\tau)$:

$$\boxed{e(\tau) = \frac{w_A L_A}{w_B L_B}\cdot\frac{1}{1+(1-m)\tau}} \qquad \left.\frac{d\log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_1^*}{D_2}\right|_{\eta=1}, \tag{34}$$

strictly decreasing in τ globally: the linearized result (32) is exact-in-sign at all tariff levels in the Cobb–Douglas case. Note that $\eta=1$ pins the expenditure shares at the *weights* $(\alpha_D, 1-\alpha_D)$ but does not restrict the weights themselves: Cobb–Douglas means unit elasticity, not equal shares. Equal home and import spending, $\alpha_D=1/2$, is a further special case.

5.3 The Marshall–Lerner condition

At $N = 2$ the multilateral condition (30) is one inequality, $J < 0$. This subsection is not a new assumption on top of it: it restates that inequality in the classical elasticity form in which the conventional wisdom is usually asserted, and then shows that in the nested model the elasticity condition and the denominator D_2 are the same object.

Reduced form: export volume $X(p^*)$ with $p^* = 1/e$, import volume $M(p)$ with $p = (1+\tau)e$, elasticities $\varepsilon_X \equiv -X'p^*/X \geq 0$, $\varepsilon_M \equiv -M'p/M \geq 0$. The balance in A's currency at producer prices:

$$\mathrm{TB}_A(e, \tau) = X(1/e) - e\, M\big((1+\tau)e\big). \tag{35}$$

Differentiate at $\tau = 0$, balanced trade ($X = eM$):

$$\frac{\partial \mathrm{TB}_A}{\partial e} = \underbrace{\frac{\varepsilon_X X}{e}}_{\text{export volume}} + \underbrace{\varepsilon_M M}_{\text{import volume}} - \underbrace{M}_{\text{valuation}} = M\big(\varepsilon_X + \varepsilon_M - 1\big). \tag{36}$$

A depreciation improves the balance iff $\varepsilon_X + \varepsilon_M > 1$: the volume responses must offset the one-for-one valuation loss. The tariff response at fixed e:

$$\frac{\partial \mathrm{TB}_A}{\partial \tau} = \frac{\varepsilon_M}{1+\tau}\, eM > 0 \quad \text{whenever } \varepsilon_M > 0. \tag{37}$$

Equilibrium $\mathrm{TB}_A(e(\tau), \tau) = 0$ and the implicit function theorem give

$$\frac{de}{d\tau} = -\frac{\partial \mathrm{TB}_A/\partial \tau}{\partial \mathrm{TB}_A/\partial e}: \tag{38}$$

the tariff creates a surplus at the initial rate (37); the rate must move in the direction that worsens the balance; under Marshall–Lerner that direction is appreciation. The condition is necessary and sufficient because $\operatorname{sign}(de/d\tau) = -\operatorname{sign}(\varepsilon_X + \varepsilon_M - 1)$.

In the nested model the condition is not free-standing. Comparing (36) with the linearized J at $N = 2$ (per unit of import value m):

$$\varepsilon_X + \varepsilon_M - 1 = D_2 = 2\alpha_D(\eta - 1) + 1, \qquad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1: \tag{39}$$

the denominator D_2 *is* the Marshall–Lerner margin, so (30) at $N = 2$ and $\varepsilon_X + \varepsilon_M > 1$ are the same condition. At $\eta = 1$ (Cobb–Douglas), $\varepsilon_X = \varepsilon_M = 1$ and the margin is exactly 1.

5.4 Quantities

From (9)–(11), with $\nu_B = d\log e_{AB}$:

$$d\log \omega_{AB} = \nu_B, \qquad d\log \mathrm{ToT}_{AB} = -\nu_B, \qquad d\log q_{AB} = (1 - 2m)\,\nu_B \;-\; m\, d\tau. \tag{40}$$

The real-rate expression is the $N = 2$ case of (67) below: the tariff appreciates A in real terms directly (it raises A's consumer prices at any e), and nominal pass-through is attenuated by the two countries' tradable exposures. A tariff therefore appreciates the currency, improves the terms of trade, raises the relative wage, and appreciates the real exchange rate: at $N = 2$ every definition of "the exchange rate" moves together.

6 Three countries

6.1 The linear system

Set $N = 3$, symmetric weights $\beta_{ij} = 1/2$, equal sizes, and perturb by $d\tau = dt_{AB}$. Baseline shares $s_{ii} = \alpha_T\alpha_D$, $s_{ij} = m/2$. From (19)–(20):

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C - \nu_B, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B - \nu_C, \tag{41}$$

(own-currency), and $d\log I_A = \frac{m}{2}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$ (common currency). Substituting into (24) for $i = B, C$ and collecting terms in $(\nu_B, \nu_C, d\tau)$ yields, with $D \equiv D_3 = 3\alpha_D(\eta-1)+\rho+1$,

$$J = -\frac{mD}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}. \tag{42}$$

The elasticities enter J only through the scalar D; the matrix part is pure counting. Inverting with $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1}F\,d\tau = -\frac{1}{3D}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D}\begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \tag{43}$$

Result 6.1 (Threshold)**.** *At the symmetric point, with $D > 0$ (the multilateral Marshall–Lerner condition* (30)*, to which all its characterizations collapse here; it holds for all $\eta \geq 1$),*

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D} < 0, \qquad \boxed{\frac{d\log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D}} \qquad \frac{d\log e_{BC}}{d\tau} = \frac{2\rho}{3D} > 0. \tag{44}$$

A appreciates against the target B for all parameters; A depreciates against the untariffed C iff $\rho > \rho_3^* \equiv 3\rho_1^*$; B depreciates against C for all parameters.

In the sign rule (31), $(-J)^{-1} = \frac{4}{3mD}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and

$$\frac{d\nu_C}{d\tau} \propto F_B + 2F_C \propto \rho - 3\rho_1^* : \tag{45}$$

the threshold is the point where C's own diversion gain ($F_C > 0$ iff $\rho > \rho_1^*$) outweighs its exposure to B's loss ($F_B < 0$).

6.2 Impossibility under a single elasticity

Impose $\rho = \eta = \sigma$. Then

$$\rho_3^* - \sigma = 3\rho_1^*\big|_{\eta=\sigma} - \sigma = \sigma(3\alpha_D - 1) + 3(1-\alpha_D)(1-\alpha_T). \tag{46}$$

Result 6.2 (Impossibility)**.** *If $\alpha_D \geq 1/3$, then $\rho_3^* > \sigma$ for every $\sigma > 0$: with one elasticity, the reversal against C is unreachable. The bound involves only α_D; α_T is unrestricted. As an illustration, at the point $\alpha_D = 1/3$, $\alpha_T = 3/4$ — equal Cobb–Douglas weights $1/4$ on each of the four goods (three tradables and the non-tradable), the paper's original calibration, not a requirement of the result — $\rho_3^* - \sigma = \frac{1}{2}$, $D = 2\sigma$, and*

$$\frac{d\log e_{AC}}{d\tau} = \frac{\sigma - \rho_3^*}{3D} = -\frac{1}{12\sigma} \longrightarrow 0^-. \tag{47}$$

6.3 Bilateral versus effective: the NEER

Decompose the solution (43) into an aggregate and a relative component:

$$\underbrace{\frac{\nu_B + \nu_C}{2}}_{\text{NEER of } A} = -\frac{\rho_1^*}{D}\, d\tau, \qquad \underbrace{\nu_B - \nu_C}_{\text{relative}} = -\frac{2\rho}{3D}\, d\tau. \tag{48}$$

The cross-origin elasticity ρ cancels from the aggregate component and fully controls the relative one.

Result 6.3 (NEER preservation)**.** *With symmetric weights, $d\nu_A^E/d\tau = -\rho_1^*/D < 0$ for all ρ: a unilateral tariff always appreciates the tariffing country's effective exchange rate, even when $\rho > \rho_3^*$ reverses the bilateral rate against C. The conventional wisdom fails bilaterally but survives in aggregate, and by* (33) *the aggregate sign is unconditional.*

Where the conditions sit empirically. Given (30), $D > 0$ and $\rho_1^* > 0$ hold at every admissible calibration, so the signs of e_{AB}, e_{BC}, and the NEER in this section are parameter-free. The one genuinely ambiguous object is the bilateral rate against the bystander: at the benchmark, $\rho_3^* = 3.96$ sits inside the range of estimated cross-origin trade elasticities (roughly 2–8: macro estimates at the low end, trade-level estimates at the high end), so either sign of $d\log e_{AC}$ is empirically plausible. By contrast the war threshold below, $\rho_1^* = 1.32$, lies beneath essentially all estimates, so the war predictions are clear-cut.

6.4 Real exchange rates

From (11) and (8), $d\log q_{Aj} = \nu_j + \alpha_T\big(d\log P_{T_j}^{\text{own}} - d\log P_{T_A}^{\text{own}}\big)$ with own-currency indices (20). Both bilateral real rates take one form (the $N = 3$ case of (67)):

$$d\log q_{Aj} = \left(1 - \tfrac{3m}{2}\right)\nu_j \;-\; \tfrac{m}{2}\, d\tau, \qquad j = B, C. \tag{49}$$

Setting $d\log q_{AC} = 0$ with ν_C from (43): $(2-3m)(\rho - 3\rho_1^*) = 3mD$, linear in ρ, so

$$\boxed{\rho_q^* = \frac{(2-3m)\,3\rho_1^* + 3m\big[\,3\alpha_D(\eta-1)+1\,\big]}{2(1-3m)}} \qquad (\rho_q^* > \rho_3^* \text{ for } m \in (0, \tfrac{1}{3})). \tag{50}$$

At the benchmark ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, $m = 0.08$): $\rho_3^* = 3.96$, $\rho_q^* = 4.93$. Averaging (49) over j gives the REER:

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \tfrac{3m}{2}\Big)\frac{\rho_1^*}{D} - \frac{m}{2} \;<\; 0 \qquad \text{for all } \rho \ (m < \tfrac{2}{3}) : \tag{51}$$

real effective appreciation is unconditional and strictly stronger than nominal, because the tariff also raises A's consumer prices directly.

6.5 Tariff configurations by superposition

Responses are linear in the tariff vector: any configuration is a sum of directed bilateral shocks, with mirror shocks obtained by relabeling (43).

$$\text{broad tariff (on } B \text{ and } C\text{):} \qquad \frac{d\log e_{Aj}}{d\tau} = \nu_B + \nu_C = -\frac{2\rho_1^*}{D} < 0, \quad j = B, C, \tag{52}$$

$$\text{symmetric war (with } B\text{):} \qquad \frac{d\log e_{AB}}{d\tau}\bigg|^w = 0, \qquad \frac{d\log e_{AC}}{d\tau}\bigg|^w = \frac{\rho - \rho_1^*}{D}. \tag{53}$$

In (52), ρ drops out entirely: uniform protection is non-discriminatory, so there is no cross-origin margin and every bilateral rate (hence the NEER) appreciates. In (53), the belligerents' bilateral component cancels and the bystander margin decides everything:

$$\frac{d\nu_A^E}{d\tau}\bigg|^w = \frac{1}{2}\,\frac{\rho - \rho_1^*}{D} > 0 \quad \Longleftrightarrow \quad \rho > \rho_1^*. \tag{54}$$

Result 6.4 (Configuration dependence)**.** *The unilateral tariff appreciates the NEER for all ρ; the reciprocal war depreciates both belligerents' NEERs iff $\rho > \rho_1^*$. Retaliation, not protection itself, is what can weaken the tariffing country's currency in aggregate.*

7 General symmetric N

7.1 The reduced system

Symmetric weights $b_{ij} = 1/(N-1)$, equal sizes, $d\tau = dt_{AB}$; the $N-2$ bystanders are identical with common ν_C. From (19)–(20):

$$d\log P_{M_A} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \qquad d\log P_{M_B} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \qquad d\log P_{M_C} = \frac{\nu_B - 2\nu_C}{N-1}, \tag{55}$$

and $d\log I_A = \frac{m}{N-1}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$. Substituting into (24) for B and a representative bystander and collecting terms (each row scaled by $(N-1)/m$; verified symbolically):

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \qquad D_N = N\alpha_D(\eta-1) + (N-2)\rho + 1. \tag{56}$$

As at $N = 3$, the elasticities enter the adjustment matrix only through the scalar D_N; the matrix is pure counting, with determinant N. Inverting:

$$\frac{d\log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{N\,D_N} \;<\; 0, \tag{57}$$

$$\frac{d\log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N\,D_N}, \tag{58}$$

$$\frac{d\log e_{BC}}{d\tau} = \frac{(N-1)\,\rho}{N\,D_N} \;>\; 0. \tag{59}$$

At $N = 3$ these are (44); at $N = 2$, (57) is (32) (with ρ absent, $D_2 = 2\alpha_D(\eta-1)+1$).

Result 7.1 (N-country threshold)**.** *A depreciates against each untariffed bystander iff*

$$\boxed{\rho > \rho_N^* = N\rho_1^*} \tag{60}$$

The threshold is linear in N: diverted expenditure splits across $N-2$ bystanders, each receiving weight $1/(N-1)$, so a larger ρ is needed to reverse any one bilateral rate. The bilateral reversal is dimension-dependent.

7.2 The NEER: dimension- and ρ-invariant sign

Averaging with symmetric weights:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \frac{-\big[N\rho_1^* + (N-2)\rho\big] + (N-2)\big[\rho - N\rho_1^*\big]}{(N-1)\,N\,D_N} = \boxed{-\frac{\rho_1^*}{D_N}} \tag{61}$$

The ρ terms cancel exactly, and N affects only the magnitude through D_N. The companion relative component is pure ρ:

$$\nu_B - \nu_C = -\frac{(N-1)\rho}{N D_N}\,d\tau. \tag{62}$$

Result 7.2 (Aggregation)**.** *In the symmetric model, for every $N \geq 2$ and every ρ, a unilateral tariff appreciates the tariffing country's NEER: $d\nu_A^E/d\tau = -\rho_1^*/D_N < 0$ by* (33)*. Cross-origin substitution reallocates the adjustment across partners* (62) *and can reverse individual bilateral rates* (58)*, but never the aggregate.*

7.3 Impossibility at general N

With $\rho = \eta = \sigma$:

$$\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \ > \ 0 \quad \text{whenever } \alpha_D \geq 1/N. \tag{63}$$

The home-bias bound weakens with N: any calibration with home share at least $1/N$ keeps the bilateral reversal invisible to single-elasticity models.

7.4 Configurations

By superposition, as in Section 6.5:

$$\text{broad tariff (on all } N-1 \text{ partners):} \qquad \frac{d\log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} \quad \forall j \ = \ \frac{d\nu_A^E}{d\tau}, \tag{64}$$

$$\text{symmetric war with } B: \qquad \left.\frac{d\log e_{AB}}{d\tau}\right|^w = 0, \qquad \left.\frac{d\log e_{AC}}{d\tau}\right|^w = \frac{\rho-\rho_1^*}{D_N}, \tag{65}$$

and the war NEER, $\frac{N-2}{N-1}\frac{\rho-\rho_1^*}{D_N}$, has the sign of $\rho - \rho_1^*$.

Result 7.3 (War threshold is independent of N)**.** *In a symmetric war, every bystander appreciates against both belligerents — and both belligerents' NEERs depreciate — iff*

$$\boxed{\rho > \rho_1^* = \frac{\rho_N^*}{N}} \tag{66}$$

for any N. With e_{AB} pinned at zero by symmetry, each bystander's equation decouples and its sign is its own impact effect, which does not dilute with N. At $\eta = 1$, $\rho_1^* = 1 - m < 1$: any $\rho \geq 1$ suffices.

7.5 Real rates at general N

Both bilateral real rates take one form (verified symbolically; contains (40) and (49)):

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big)\nu_j \ - \ \frac{m}{N-1}\,d\tau, \qquad j = B, C. \tag{67}$$

Setting $d\log q_{AC} = 0$ with (58) gives the real bilateral threshold (for $m < 1/N$):

$$\boxed{\rho_q^*(N) = \frac{\big[(N-1) - mN\big]\,N\rho_1^* + mN\big[N\alpha_D(\eta-1)+1\big]}{(N-1)(1-mN)}} \tag{68}$$

($\rho_q^*(3) = 4.93$, $\rho_q^*(4) = 7.34$, $\rho_q^*(5) = 10.40$ at the benchmark), and averaging (67) gives the REER:

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \frac{mN}{N-1}\Big)\frac{\rho_1^*}{D_N} - \frac{m}{N-1} \ < \ 0 \qquad \text{for all } \rho, N \ \big(m < \tfrac{N-1}{N}\big). \tag{69}$$

7.6 Correspondence across N

object	$N = 2$	$N = 3$	general N
ML condition (30)	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	$-J$ a nonsingular M-matrix
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 2)
sign rule	one impact, unconditional	weights $\frac{1}{3}\left(\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right)$	sign $\left(\sum_j \omega_{ij} F_j\right)$, $\omega \geq 0$
bilateral threshold	none (no bystander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree-$(N-1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$ (50)	$\rho_q^*(N)$ (68)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$: sign invariant
war threshold	—	ρ_1^*	ρ_1^*, independent of N

8 Asymmetry

8.1 General baselines

Nothing in Section 4 used symmetry: at any balanced baseline, the system (19)–(24) holds with the observed shares (b_{ij}, h_i) and incomes y_i as coefficients. The paper's appendix solves it for $N = 3$ at a general baseline (unequal shares and sizes, existing tariffs). The bilateral slope against the bystander then has denominator $\det(-J) > 0$ and numerator quadratic in ρ:

$$\operatorname{sign} \frac{d \log e_{AC}}{d\tau} = \operatorname{sign} \mathcal{N}(\rho), \qquad \mathcal{N}(\rho) = c_2\,\rho^2 + c_1\,\rho + c_0, \qquad c_2 = b_{AB}\, b_{AC}\, b_{CA}\, b_{CB}\,(1-h_A)(1-h_C)\, y_A\, y_C. \tag{70}$$

Two consequences:

- $c_2 > 0$ whenever A and C each buy from both foreign origins (every share in the product is positive). Then $\mathcal{N}(\rho) > 0$ for ρ large enough: sufficiently strong cross-origin substitution reverses e_{AC} at *any* such baseline. Asymmetry moves the threshold; it cannot remove the reversal region.

- The threshold itself is the largest root of $\mathcal{N}(\rho) = 0$. At the symmetric free-trade baseline that root is $3\rho_1^*$, recovering (44).

8.2 Asymmetric exposure to the target: the κ parameterization

To compare asymmetry across N, let the target B carry a fixed multiple κ of the symmetric bilateral share, with the weight matrix kept symmetric (hence balanced):

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \qquad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \qquad b_{CC'} = \frac{1 - 2\,b_{CA}}{N-3}. \tag{71}$$

$\kappa = 1$ is exact symmetry; $\kappa \in (0, N-1]$ scales the target's importance to A while remaining comparable as N changes. The bystanders are still interchangeable, so the reduced system again has two unknowns (ν_B, ν_C), and it solves in closed form (symbolically; replication package): with NEER weights $w_{Aj} = b_{Aj}$, the response $d\nu_A^E/d\tau$ is an explicit rational function of (ρ, N, κ) that collapses to $-\rho_1^*/D_N$ exactly at $\kappa = 1$. Its numerator is $-\kappa\,\rho_1^*$ times a quadratic $Q(\rho)$ with leading coefficient proportional to $(\kappa - 1)(N - 1 - \kappa)$.

Result 8.1 (Asymmetric NEER). *Holding κ fixed as $N \to \infty$,*

$$N\,\frac{d\nu_A^E}{d\tau} \;\longrightarrow\; -\,\kappa\,\frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} \;=\; -\,\kappa \lim_{N\to\infty} \frac{N\rho_1^*}{D_N}\;: \tag{72}$$

to leading order, asymmetric exposure scales the symmetric NEER response by the target's relative importance κ and preserves its sign.

At finite N the sign of the exact solution follows from Q, and the direction of the asymmetry matters:

- $\kappa \geq 1$ (a disproportionately *large* target): the coefficients of Q are positive for large N (the leading coefficient by $(\kappa-1)(N-1-\kappa) \geq 0$, the others with leading terms N^2 and $N^2\alpha_D(\eta-1)$), so $d\nu_A^E/d\tau < 0$ for every ρ: effective appreciation survives.
- $\kappa < 1$ (a disproportionately *small* target): Q has a single positive root,

$$\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1-\kappa}\big(1 + O(N^{-1})\big), \tag{73}$$

  beyond which the NEER *depreciates*: with little weight on the target, the depreciations against the many bystanders outweigh the appreciation against B in the average. As $\kappa \to 0$ the flip point approaches the bilateral threshold $N\rho_1^*$ — with a weightless target, the NEER simply *is* the bystander rates.

The exact cancellation of (61) is therefore one-sided. Tariffs on disproportionately large partners preserve the conventional aggregate sign for every ρ; tariffs on minor partners can reverse it, but only at ρ of order N — far above the war threshold ρ_1^* and, at moderate N, above estimated elasticities ($\rho_E^* \approx 22$ at $N = 10$, $\kappa = 0.5$, benchmark parameters).

9 Canonical PTA model: competing exporters

9.1 Setup

Countries A (importer), B and C (exporters). Two goods: x and numeraire y. Endowments: A holds only y; B and C hold x (and possibly y). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \qquad u_i' > 0,\ u_i'' < 0. \tag{74}$$

A levies specific tariffs t_B, t_C on imports of x from B and C. Let q_j be the price received by exporter j (the world price of its good) and p the domestic price of x in A. Both origins sell in A at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \tag{75}$$

9.2 Demands and market clearing

From $u_i'(x_i) =$ (local price of x):

$$M_A(p) = x_A^d(p), \qquad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \tag{76}$$

with $M_A' < 0$ and $E_j' > 0$. Market clearing for x:

$$M_A(p) = E_B(p - t_B) + E_C(p - t_C). \tag{77}$$

Equation (77) determines p given (t_B, t_C); then q_B, q_C follow from (75).

9.3 Comparative statics of a discriminatory tariff

Totally differentiate (77) with respect to t_B, holding t_C:

$$M_A' \frac{dp}{dt_B} = E_B'\Big(\frac{dp}{dt_B} - 1\Big) + E_C' \frac{dp}{dt_B} \quad\Longrightarrow\quad \frac{dp}{dt_B} = \frac{E_B'}{E_B' + E_C' - M_A'} \in (0,1). \tag{78}$$

Therefore

$$\frac{dq_B}{dt_B} = -\frac{E_C' - M_A'}{E_B' + E_C' - M_A'} \ < \ 0, \qquad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} \ > \ 0. \tag{79}$$

Result 9.1 (Viner–Mundell)**.** *A discriminatory tariff on B worsens B's terms of trade and improves C's terms of trade. Equivalently, a preferential tariff cut toward B (lower t_B) improves B's and worsens C's terms of trade.*

9.4 The PTA assumptions as restrictions on the three-country threshold

An improvement in C's terms of trade is a rise in e_{AC} (10). The canonical model signs this unconditionally from $E_j' > 0$ and $M_A' < 0$; Section 6 requires $\rho > 3\rho_1^*$. The difference is the three pieces of the threshold (benchmark values: $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$):

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B\text{–}C\text{ linkage}} + \underbrace{3\alpha_D(\eta - 1)}_{\text{home switching}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\text{openness credit}} = 3.0 + 1.2 - 0.24 = 3.96. \qquad (80)$$

The linkage claim collects the two forces running through B–C trade: B's income loss cuts its purchases from C, and C substitutes toward B's cheapened goods. Each canonical assumption removes one piece. Recomputed thresholds under each restriction (severing trade changes the baseline, so the pieces are approximately additive):

restriction	threshold ρ_3^*	piece removed
none (benchmark)	3.96	—
no B–C trade ($\beta_{BC} = \beta_{CB} = 0$)	1.05	the linkage claim
no home tradables ($\alpha_D = 0$)	1.80	the home-switching claim
no non-tradables ($\alpha_T = 1$)	3.60	credit maximized (small alone)
$\alpha_D = 0$ and $\alpha_T = 1$	none: reversal for all $\rho > 0$	
no B–C trade and $\alpha_D = 0$	none: reversal for all $\rho > 0$	

Three readings:

- The largest piece is the B–C linkage: severing it alone moves the threshold from 3.96 to 1.05.
- Non-tradables are not an independent force: with the linkage severed and $\alpha_D = 0$, no threshold exists at any α_T. The non-tradable share enters only through the credit.
- Two routes reach $\rho_3^* \le 0$: $\{\alpha_D = 0,\ \alpha_T = 1\}$ or $\{\alpha_D = 0,\ \beta_{BC} = \beta_{CB} = 0\}$. The canonical model takes both, adds quasilinearity, and in the Viner case sets $\rho = \infty$.

Result 9.2. *The canonical PTA model is the restriction of the three-country model to a subspace of structures on which $\rho_3^* \le 0$. The unconditional Viner–Mundell sign is the threshold condition of Section 6 evaluated where it cannot bind.*

The analytical literature maps onto the table:

- competing exporters (Bagwell–Staiger 1999; Ornelas 2005): $\beta_{BC} = \beta_{CB} = 0$, quasilinear;
- symmetric endowment blocs (Krugman 1991; Bond–Syropoulos 1996): $\alpha_D \approx 0$, $\alpha_T = 1$, one elasticity;
- Viner: $\rho = \infty$;
- Kemp–Wan 1976: none of these restrictions, and no sign claims.

The applied literature sits on the other side of the same divide. Large simulation models — computable general equilibrium models of trade agreements in the GTAP tradition, and modern quantitative trade models — contain every force in (80): they use a two-level import nest like (5)–(6), full trade matrices, non-tradables, and income effects. In their simulations, a discriminatory tariff sometimes improves and sometimes worsens the excluded country's terms of trade, depending on the configuration. That is what crossing a threshold looks like when the threshold has not been written down: the models produce both signs, but no statement of when each occurs.

The division of labor is then simple. The analytical PTA models prove the sign by deleting the opposing forces. The applied models keep all the forces but report numbers, not conditions. The condition itself is $\rho > 3\rho_1^*$. And because the bilateral threshold ($N\rho_1^*$), the war threshold (ρ_1^*), and the NEER response ($-\rho_1^*/D_N$) are all built from the same object, the three forces in (80) — the B–C linkage, home expenditure switching, and the openness credit — are the complete list of what any of these models is implicitly weighing when it signs an exchange-rate or terms-of-trade response.